

Timestamp & Validation-Based Protocols — DBMS

Definition: Timestamp Ordering Protocols ensure conflict serializability without using locks by assigning a unique monotonic timestamp **TS(T_i)** to each transaction upon start and enforcing operation ordering according to timestamp sequence.

1. Detailed Technical Explanation

1. System Timestamps

Each data item **X** maintains two timestamps:

- **W-TS(X)** : Largest timestamp of any transaction that executed **Write(X)** .
- **R-TS(X)** : Largest timestamp of any transaction that executed **Read(X)** .

2. Basic Timestamp Ordering Rules

Read Operation **Read(X)** by Transaction **T_i**:

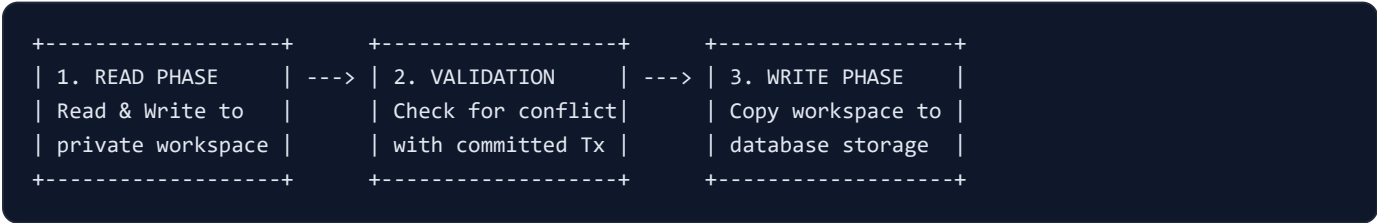
1. If **TS(T_i) < W-TS(X)**: **T_i** is attempting to read an overwritten value. **Abort and Rollback** **T_i**.
2. Else: Execute **Read(X)** and update **R-TS(X) = \max(R-TS(X), TS(T_i))**.

Write Operation **Write(X)** by Transaction **T_i**:

1. If **TS(T_i) < R-TS(X)**: **T_i** is attempting to write a value needed by a younger transaction. **Abort and Rollback** **T_i**.
2. If **TS(T_i) < W-TS(X)**: **T_i** is writing an obsolete value (**Thomas Write Rule**: Ignore write and continue).
3. Else: Execute **Write(X)** and update **W-TS(X) = TS(T_i)**.

2. Validation-Based (Optimistic) Protocol

Used when read operations dominate write operations (low conflict rate). Execution is split into 3 phases:



Validation Test Condition:

For all **T_k** committed before **T_i**:

- **Finish(T_k) < StartValidation(T_i)**, OR
- **Writeset(T_k) \cap Readset(T_i) = \emptysetset**.

3. Core Concepts & Memory Keywords

- **Monotonic Timestamps:** Unique growing values (System clock or global counter).
 - **Thomas Write Rule:** Optimization ignoring outdated write requests rather than aborting.
 - **Optimistic Concurrency Control:** Read -> Validate -> Write phases.
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4. Must-Write Points for Exams

- Timestamp ordering protocols guarantee **freedom from deadlocks** because transactions are never made to wait indefinitely.
 - Starvation can occur under timestamp ordering if a long transaction is repeatedly aborted by younger transactions.
 - Validation protocols perform all modifications on local workspaces before validating.
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5. Quick Recall Flow

Compare $TS(T_i)$ with $W-TS(X)/R-TS(X)$ -> Valid? Proceed : Outdated? Abort & Restart with New Timestamp